Robot Vision: Mock Exam 2 (Solutions)

Chapters 2 to 12 · 90 minutes · no documents · 40 points

Part A: Multiple choice

22 points, about 35 min

A1 Which statement correctly describes a fully connected layer? (ch. 2)

(a) All units share one weight vector.
(b) Each unit sees only a small local region of the input.
(c) ✓ **Every input is connected to every unit, and each unit has its own weights and bias.**
(d) Each input is connected to exactly one unit.

Sharing weights is the convolution, not the FC layer.

A2 Why is ordinary K-class classification unsuitable for ordinal regression (e.g. estimating an age)? (ch. 2)

(a) ✓ **It does not distinguish a prediction close to the correct class from one far from it.**
(b) It cannot be trained with gradient descent.
(c) It requires the classes to be independent images.
(d) It cannot handle more than ten classes.

Predicting 31 for 32 should cost less than predicting 50.

A3 What are the standard roles of validation data and test data? (ch. 3)

(a) Validation data are used only after the test.
(b) ✓ **Validation data select hyperparameters; test data are kept for the final evaluation only.**
(c) Both are used to fit the weights.
(d) Test data select hyperparameters; validation data fit the weights.

Never tune on the test set.

A4 What is the main idea of dropout? (ch. 3)

(a) Switch off units only at test time.
(b) ✓ **Randomly switch off units during training to reduce overfitting; all units are used at test time.**
(c) Permanently delete the smallest weights.
(d) Add noise to the labels.

Survivors are scaled by $1/(1-p)$.

A5 What does invariance mean for a visual recognition feature? (ch. 4)

(a) The feature is the same for all images.
(b) The feature distinguishes very similar categories.
(c) ✓ **The feature stays roughly unchanged under variations irrelevant to the category (pose, lighting, position).**
(d) The feature changes strongly when the object moves.

Sensitivity is the other requirement: tell similar classes apart.

A6 A $64 \times 64 \times 3$ image goes through 16 filters 3×3 (stride 1, padding 1), ReLU, then 2×2 max pooling with stride 2. Output size? (ch. 4)

(a) $32 \times 32 \times 3$
(b) $64 \times 64 \times 16$
(c) ✓ $32 \times 32 \times 16$
(d) $16 \times 16 \times 16$

Conv keeps 64 (padding 1), pooling halves it; 16 filters = 16 channels.

A7 Why does a ResNet bottleneck block start with a 1×1 convolution? (ch. 5)

(a) ✓ **To reduce the number of channels, so the costly 3×3 convolution runs on fewer channels.**

(b) To replace the residual connection.
(c) To apply pooling without weights.
(d) To increase the spatial resolution.

The last 1×1 restores the channels.

A8 Why does SSD predict from feature maps of several resolutions? (ch. 5)

(a) Because each map detects one class.
(b) To make the network faster than a single map.
(c) ✓ **To handle objects of different sizes: fine maps for small objects, coarse maps for large ones.**
(d) To remove the need for NMS.

Multi-scale detection.

A9 In activation maximization, why is a regularizer $R(x)$ added? (ch. 6)

(a) To make the optimization faster.
(b) ✓ **To keep x looking like a natural image rather than high-frequency noise.**
(c) To keep the network weights fixed.
(d) To reduce the number of classes.

The result also depends on the initialization.

A10 How does SmoothGrad reduce the noise of saliency maps? (ch. 6)

(a) By using only the first layer.
(b) By randomizing the weights.
(c) By blurring the input image once.
(d) ✓ **By averaging the maps computed on several noisy copies of the input.**

Average over noise.

A11 How does Mixup build a new training example? (ch. 7)

(a) It searches augmentations with reinforcement learning.
(b) It cuts a square out of an image.
(c) It mixes the inputs but keeps the label of the first image.
(d) ✓ **It mixes two inputs and their labels with the same coefficient.**

Weights 0.4/0.6 give the soft label 0.4/0.6.

A12 What does consistency regularization require on an unlabeled image? (ch. 7)

(a) ✓ **Similar predictions before and after a suitable transformation.**
(b) That the image is removed from training.
(c) A different prediction after each transformation.
(d) A manual label.

Unlabeled data help this way.

A13 What does the triplet loss require? (ch. 8)

(a) All three samples must have the same feature.
(b) The positive must be farther than the negative.
(c) The anchor must be closer to the negative.
(d) ✓ **The anchor must be closer to the positive than to the negative by at least a margin.**

$d(a, n) \geq d(a, p) + m$.

A14 How does FGSM build an untargeted adversarial example? (ch. 8)

(a) It adds random Gaussian noise.
(b) It moves the input to decrease the loss of the true class.
(c) ✓ **It adds a small step along the sign of the gradient of the loss with respect to the input.**
(d) It retrains the model on noisy images.

It increases the loss of the true class.

A15 How is an RNN trained with backpropagation through time (BPTT)? (ch. 9)

(a) Only the last step is trained.
(b) Gradients are not used.
(c) Each time step has its own independent weights.
(d) ✓ **The RNN is unfolded in time and gradients flow back through steps that share the same weights.**

Same weights at each step.

A16 What do the adjacency matrix A and the node-feature matrix X represent? (ch. 9)

(a) Both give the node positions in a drawing.
(b) A gives the features; X the connections.
(c) ✓ **A gives the connections between nodes; X gives the feature vector of each node.**
(d) A gives the labels of the nodes.

Node numbering and drawing positions are arbitrary.

A17 How does the original seq2seq model translate a sentence? (ch. 10)

(a) The decoder reads the source directly at each step.
(b) The encoder generates the translation.
(c) ✓ **The encoder summarizes the source in its final hidden state, and the decoder generates the translation autoregressively.**
(d) Both use self-attention only.

Fixed-length bottleneck.

A18 What is the difference between MSA and the MLP in a ViT block? (ch. 10)

(a) Neither changes the features.
(b) Both mix tokens.
(c) ✓ **MSA mixes information across tokens; the MLP transforms each token separately.**
(d) The MLP mixes tokens; MSA works token by token.

Only MSA mixes tokens.

A19 Which property do attention weights computed with a softmax have? (ch. 11)

(a) They can be negative.
(b) They are all equal.
(c) They sum to the number of regions.
(d) ✓ **They are non-negative and sum to one.**

Then a weighted sum of the region features.

A20 How can reinforcement learning (e.g. GRPO) train a reasoning model? (ch. 11)

(a) A human writes the reference reasoning for every step.
(b) The model copies the reasoning of another model.
(c) Each sentence of the reasoning is scored by hand.
(d) ✓ **Sample several answers, reward the correct final answers, and increase the probability of high-reward responses.**

Verifiable rewards on the final answer.

A21 How is an autoencoder trained? (ch. 12)

(a) ✓ **To reconstruct its input at the output, through a smaller intermediate code; no labels are needed.**
(b) To fool a discriminator.
(c) To predict the added noise.
(d) To predict the class of the input.

Squared error, or cross-entropy for binary data.

A22 Why does the VAE use the reparameterization trick? (ch. 12)

(a) To remove the KL term.
(b) ✓ **Writing $z = \mu + \sigma \odot \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$ lets gradients flow through μ and σ.**
(c) To make the decoder invertible.
(d) To compute the exact likelihood.

Sampling itself is not differentiable.

Part B: Short answers

12 points, 3 per question, about 35 min

B1 Why does multi-label classification use one sigmoid per label instead of a softmax? (ch. 2)

Several labels can be true at the same time. Softmax couples the outputs (they sum to 1), while independent sigmoids give one yes/no probability per label.

B2 What is the difference between batch normalization and layer normalization? Which one do Transformers use, and why? (ch. 3)

BN uses statistics across the samples of the mini-batch; LN uses statistics across the features of one example. Transformers use LN, which does not depend on the batch size.

B3 Why does ViT need large-scale pre-training to beat CNNs? (ch. 10)

ViT has a weak inductive bias: nothing built in says that nearby pixels are related. It must learn this from data, so with little data CNNs win; with very large pre-training ViT catches up.

B4 Compare how a GAN and a VAE are trained. (ch. 12)

GAN: a generator and a discriminator play a min-max game; no likelihood is computed. VAE: an encoder and a decoder are trained by minimizing $-$ELBO = reconstruction + KL to the prior.

Part C: Calculations

6 points, 2 per question, about 15 min

C1 A $32 \times 32 \times 3$ image goes through 8 filters 5×5 (stride 1, padding 2), then 2×2 max pooling with stride 2. Output size? (ch. 4)

Conv: $\lfloor (32 + 4 - 5)/1 \rfloor + 1 = 32$, so $32 \times 32 \times 8$. Pooling: $16 \times 16 \times 8$.

C2 With He initialization, what is the standard deviation of the weights of a 3×3 convolution with 8 input channels (ReLU)? (ch. 3)

Fan-in $M = 3 \cdot 3 \cdot 8 = 72$; std $= \sqrt{2/72} = \sqrt{1/36} = \mathbf{1/6} \approx \mathbf{0.17}$.

C3 Mixup combines a cat image and a dog image with $\lambda = 0.3$ for the cat. What are the training input and target? (ch. 7)

Input $0.3\, x_{\text{cat}} + 0.7\, x_{\text{dog}}$; soft label: **0.3 cat, 0.7 dog**.